Expectation-Maximization Algorithm (EM Algorithm)

$\mathcal{X} = \{x_i\}_{i=1}^{N}$ likelihood $\Rightarrow$ $\mathcal{L}(\Phi|\mathcal{X}) = \prod_{i=1}^{N} p(x_i|\Phi)$

loglikelihood $\Rightarrow$ $\log \mathcal{L}(\Phi|\mathcal{X}) = \sum_{i=1}^{N} \log \left[\sum_{k=1}^{K} p(x_i|C_k) Pr(C_k) \right]$

two sets of random variables

Z = cluster memberships (hidden/latent variables)

Φ = parameters $\left[\hat{P}_1, \hat{P}_2, \ldots, \hat{P}_k, \hat{\Sigma}_1, \hat{\Sigma}_2, \ldots, \hat{\Sigma}_k\right]$

E-STEP: $E\left[\mathcal{L}(\Phi|\mathcal{X}, Z) \,\middle|\, \mathcal{X}, \Phi^{(t)}\right]$

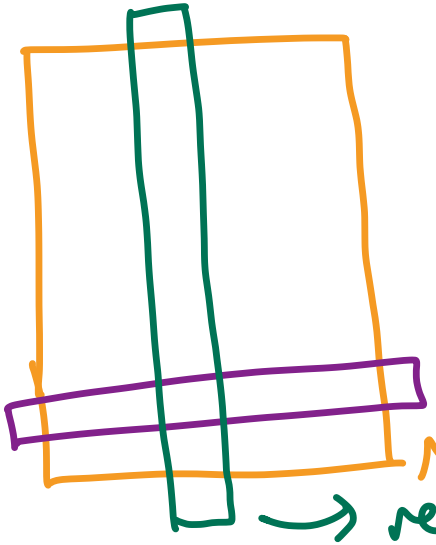
M-STEP: $\Phi^{(t+1)} = \arg\max_{\Phi} E\left[\mathcal{L}(\Phi|\mathcal{X}, Z) \,\middle|\, \mathcal{X}, \Phi^{(t)}\right]$

E-step updates:

$$h_{ik} = E[z_{ik} \mid x, \Phi^{(t)}] = \frac{p(x_i \mid C_k, \Phi^{(t)}) Pr(C_k)}{\sum_{c=1}^{K} p(x_i \mid C_c, \Phi^{(t)}) Pr(C_c)}$$

$h_{ik} \geq 0 \quad \forall (i,k)$

$\sum_{k=1}^{K} h_{ik} = 1$

$H =$ 

$N \times K$

→ each row sums up to 1.

→ relative weight of cluster k.

M-step updates:

$$\hat{Pr}^{(t+1)}(C_k) = \frac{\sum_{i=1}^{N} h_{ik}}{N}$$

$$\hat{\mu}_k^{(t+1)} = \frac{\sum_{i=1}^{N} h_{ik} \cdot x_i}{\sum_{i=1}^{N} h_{ik}}$$

$$\hat{\Sigma}_k^{(t+1)} = \frac{\sum_{i=1}^{N} h_{ik} (x_i - \hat{\mu}_k^{(t+1)})(x_i - \hat{\mu}_k^{(t+1)})^T}{\sum_{i=1}^{N} h_{ik}}$$

Hierarchical Clustering

- finding groups such that data points in a group are more similar to each other than the data points in different groups.

→ ? CLOSER ?

COMPONENT #1 : The distance between data points

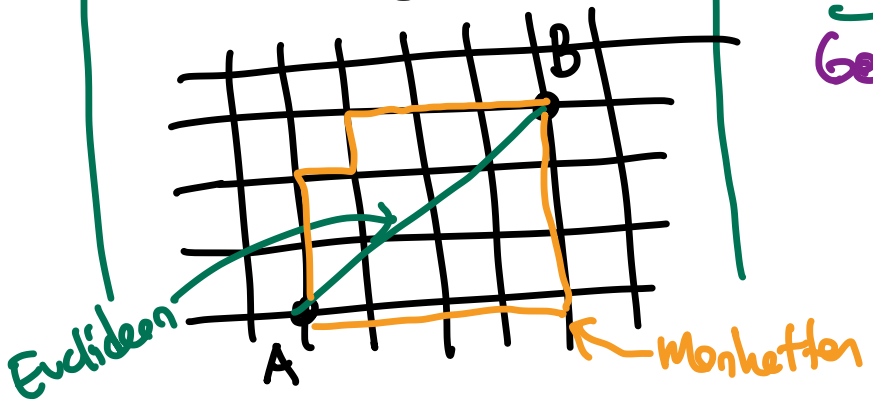
$$\text{distance} \propto \frac{1}{\text{similarity}} \quad \text{or} \quad \text{similarity} \propto \frac{1}{\text{distance}}$$

Euclidean Distance.

$$d(x_i, x_j) = \|x_i - x_j\|_2 = \sqrt{x_i^T \cdot x_i - 2 x_i^T \cdot x_j + x_j^T x_j} = \sqrt{\sum_{d=1}^{D} (x_{id} - x_{jd})^2}$$

Manhattan Distance

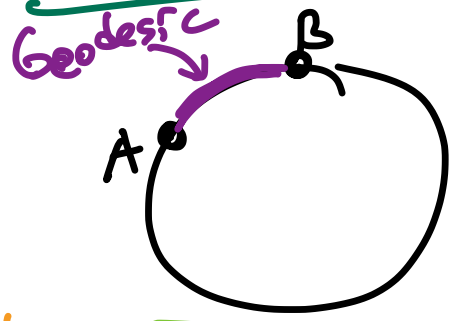
$$d(x_i, x_j) = \|x_i - x_j\|_1 = \sum_{d=1}^{D} |x_{id} - x_{jd}|$$



Mahalanobis Distance ...

$$d(x_i, x_j) = \sqrt{(x_i - x_j)^T M (x_i - x_j)}$$

Geodesic Distance



$d(a,b) \geq 0$
$d(a,b) = 0$ iff $a = b$
$d(a,b) + d(b,c) \geq d(a,c)$

COMPONENT #2: The direction to Proceed

Agglomerative
(bottom-to-top)

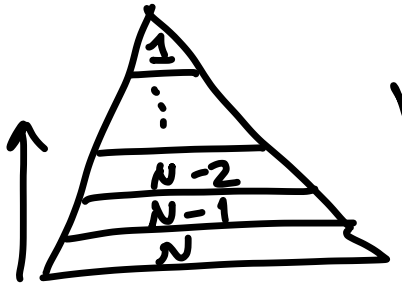
- combines small clusters into bigger ones
- starts with "N" clusters

① ② ③ ④ ⑤ ⑥

Divisive
(top-to-bottom)

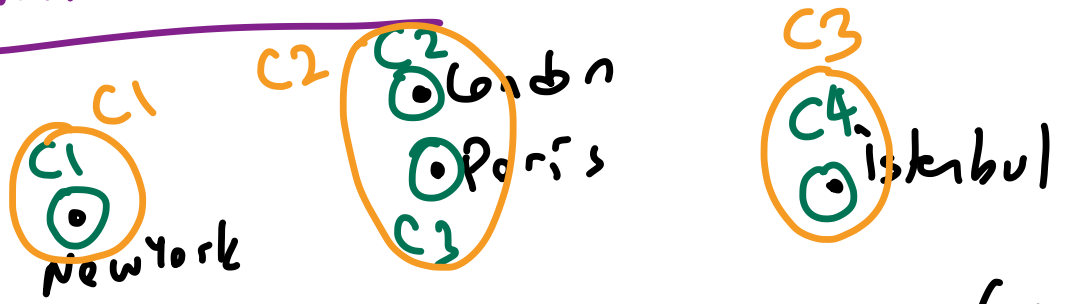
- divides big clusters into smaller ones
- starts with "1" cluster

[1, 2, 3, 4, 5, 6]



5 vs 1 $\binom{6}{1}$
4 vs 2 $\binom{6}{2}$
3 vs 3 $\binom{6}{3}$

COMPONENT #3: The distance function between groups of data



$$distance(C_1, C_2) = distance(\text{New York}, \{\text{London}, \text{Paris}\})$$

$$distance(C_2, C_3) = distance(\{\text{London}, \text{Paris}\}, \text{Istanbul})$$

Centroid clustering:

$$d(G_A, G_B) = \left\| \frac{\sum_{x_i \in G_A} x_i}{|G_A|} - \frac{\sum_{x_i \in G_B} x_i}{|G_B|} \right\|_2$$

$|G_A|$: cardinality # of members

Single-Link Clustering

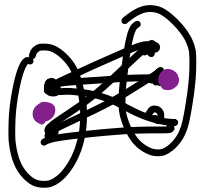
$$d(G_A, G_B) = \min_{\substack{x_i \in G_A \\ x_j \in G_B}} d(x_i, x_j)$$

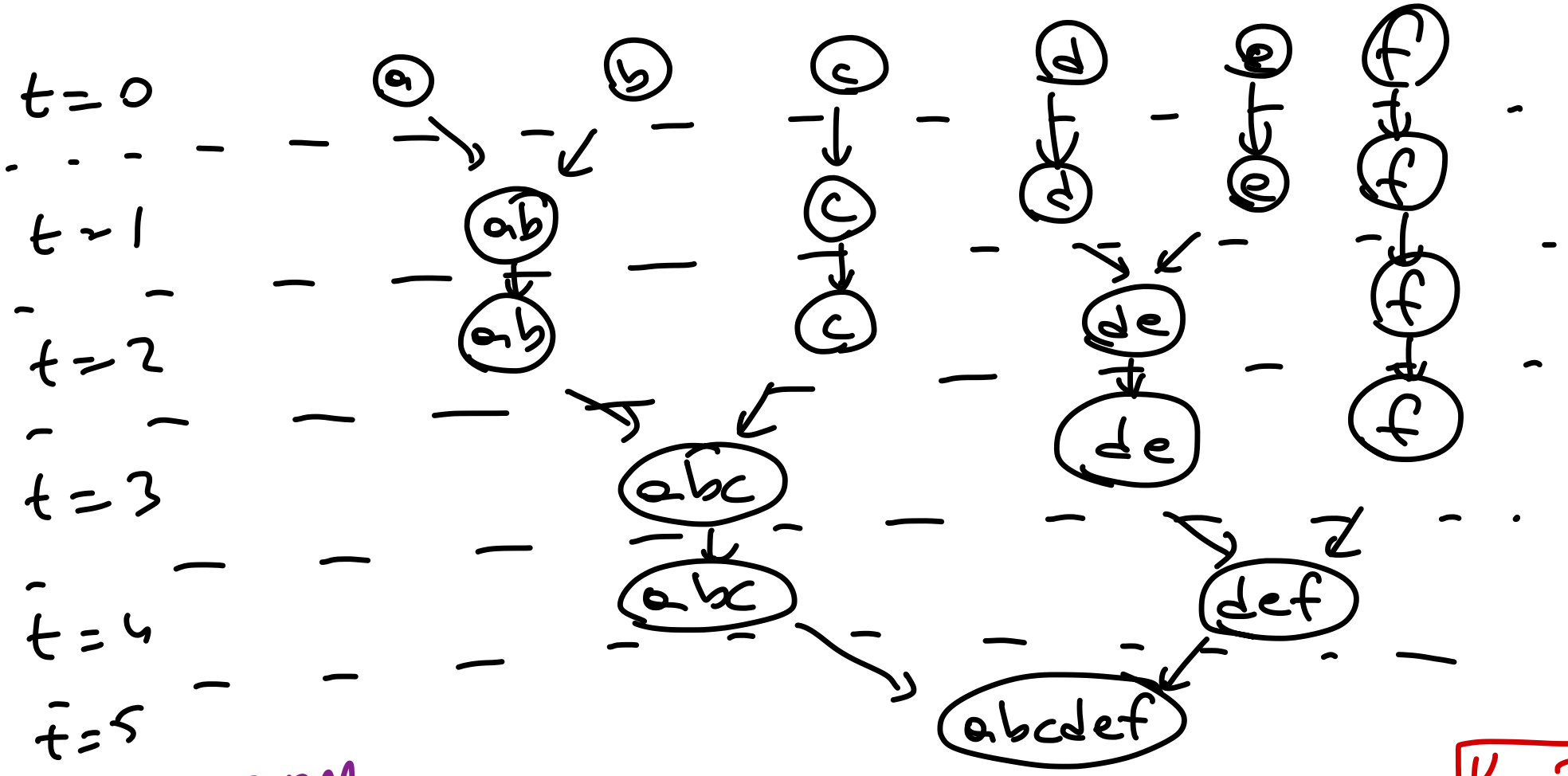
Complete-Link clustering:

$$d(G_A, G_B) = \max_{\substack{x_i \in G_A \\ x_j \in G_B}} d(x_i, x_j)$$

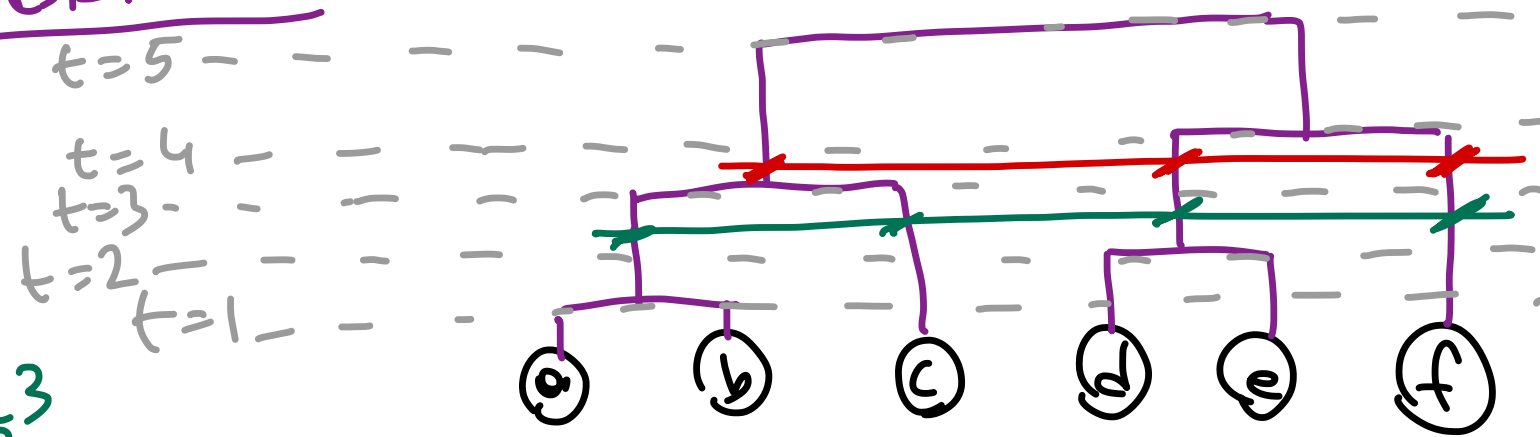
Average-Link Clustering:

$$d(G_A, G_B) = \frac{\sum_{x_i \in G_A} \sum_{x_j \in G_B} d(x_i, x_j)}{|G_A||G_B|}$$





DENDOGRAM



K=4

$C_1 = \{a, b\}$

$C_2 = \{c\}$

$C_3 = \{d, e\}$

$C_4 = \{f\}$

K=3

$C_1 = \{a, b, c\}$

$C_2 = \{d, e\}$

$C_3 = \{f\}$